

WHAT IS STATISTICS?

- ❖ Statistics is the science of collecting, organizing, analyzing, and interpreting data.
- ❖ It helps to uncover patterns and relationships in data for informed decision-making.
- ❖ Statistics is widely used in fields such as business, healthcare, engineering, and social sciences.

VARIOUS TOOLS FOR STATISTICAL ANALYSIS

- ❖ **Minitab:** A statistical software tailored for quality improvement and Six Sigma projects, offering user-friendly interfaces.
- ❖ **R:** A free, open-source programming language designed specifically for statistical computing and graphics.
- ❖ **Python:** A versatile, widely-used programming language with powerful libraries for data analysis, such as Pandas, NumPy, and SciPy.
- ❖ **SPSS:** A software package used for statistical analysis in social science, popular for surveys and academic research.
- ❖ **SAS:** A suite of statistical software used in data management, advanced analytics, and predictive modelling.

WHY EXCEL?

- ❖ Excel is the most widely available and accessible tool across various industries.
- ❖ It offers a balance of ease of use, flexibility, and powerful built-in statistical functions.
- ❖ Most professionals are already familiar with Excel, making it an ideal starting point for learning statistics.
- ❖ Excel allows for seamless integration of data entry, analysis, and visualization in a single platform.

DESCRIPTIVE VS. INFERENCE STATISTICS

- ❖ Descriptive Statistics: Summarize and describe the main features of a dataset.
- ❖ Examples: Mean, median, standard deviation, and frequency distributions.
- ❖ Inferential Statistics: Make predictions or inferences about a population based on a sample.
- ❖ Examples: Hypothesis testing, and confidence intervals.

QUALITATIVE VS. QUANTITATIVE DATA

- ❖ Qualitative Data: Non-numeric data representing categories or labels.
- ❖ Examples: Colors, gender, brands, and opinions.
- ❖ Quantitative Data: Numeric data representing measurable quantities.
- ❖ Examples: Age, income, temperature, and sales figures.

MEASUREMENT SCALES

- ❖ Nominal Scale: Categorical data without a specific order (e.g., Gender, Colors).
- ❖ Ordinal Scale: Categorical data with a meaningful order but no consistent difference between categories (e.g., Rankings, Satisfaction Levels).
- ❖ Interval Scale: Numeric data with consistent intervals but no true zero point (e.g., Temperature in Celsius).
- ❖ Ratio Scale: Numeric data with a true zero point and consistent intervals (e.g., Weight, Height, Sales).
- ❖ Understanding measurement scales is critical for selecting appropriate statistical methods.

EXCEL'S DATA ANALYSIS TOOLPAK

❖ What is the Data Analysis ToolPak?

- ❖ A powerful Excel add-in that provides advanced data analysis tools.
- ❖ It allows users to perform complex analyses, such as descriptive statistics, inferential statistics, regression, and ANOVA, with ease.

❖ Why Use the ToolPak?

- ❖ The ToolPak simplifies statistical computations without requiring complex formulas.
- ❖ It offers a user-friendly interface for generating statistical outputs quickly.

❖ How to Install the ToolPak

- ❖ Go to the File tab → Options → Add-Ins.
- ❖ Select Excel Add-Ins at the bottom and click Go.
- ❖ Check Analysis ToolPak and click OK to enable it.

WARM-UP ACTIVITY — THE INSURANCE DATASET

❖ Objective:

- ❖ Use the insurance dataset provided in the course resources to explore basic descriptive statistics and create a histogram.
- ❖ This exercise will familiarize you with using Excel for simple statistical analysis.

❖ Steps:

- ❖ Open the dataset in Excel.
- ❖ Use the Data Analysis ToolPak to:
 - ❖ Generate descriptive statistics for the charges column.
 - ❖ Create a histogram for the charges to visualize its distribution.
- ❖ Note: This is a warm-up activity to get hands-on practice; detailed explanations will follow in later sections.

INTRODUCTION TO DESCRIPTIVE STATISTICS

- ❖ Descriptive statistics provide a summary of a dataset, helping to understand its key characteristics.
- ❖ They focus on measures that describe the center, spread, and shape of the data.
- ❖ These statistics are the foundation for further analysis and interpretation of data.
- ❖ Excel offers several built-in functions to compute descriptive statistics quickly.

KEY MEASURES IN DESCRIPTIVE STATISTICS

❖ Measures of Central Tendency

- ❖ Mean: The average of all data points.
- ❖ Median: The middle value when data is ordered.
- ❖ Mode: The most frequently occurring value in the dataset.

❖ Measures of Dispersion

- ❖ Range: The difference between the maximum and minimum values.
- ❖ Variance: The average of the squared differences from the mean.
- ❖ Standard Deviation: The square root of the variance, indicating the average deviation from the mean.

ADDITIONAL DESCRIPTIVE STATISTICS

- ❖ Minimum and Maximum:
 - ❖ Provide the smallest and largest values in the dataset.
- ❖ Count and Sum
 - ❖ Provides the number of values and the sum of these in the dataset
- ❖ Quartiles:
 - ❖ Divide the dataset into four equal parts, helping to understand the distribution.
- ❖ Skewness:
 - ❖ Indicates whether the data distribution is symmetric or skewed to the left/right.
- ❖ Kurtosis:
 - ❖ Measures the "tailedness" of the data distribution, indicating how extreme values behave.

USING EXCEL TO COMPUTE DESCRIPTIVE STATISTICS

- ❖ Using Data Analysis ToolPak:
 - ❖ Go to Data → Data Analysis → Select Descriptive Statistics.
 - ❖ Input the range of data and check Summary Statistics.
 - ❖ Excel will generate the mean, median, mode, standard deviation, variance, and more in a single output.

Column1	
Mean	13270.42227
Standard Error	331.0674543
Median	9382.033
Mode	1639.5631
Standard Deviation	12110.01124
Sample Variance	146652372.2
Kurtosis	1.606298653
Skewness	1.515879658
Range	62648.55411
Minimum	1121.8739
Maximum	63770.42801
Sum	17755824.99
Count	1338

UNDERSTANDING RANGE IN EXCEL

❖ What is a Range?

- ❖ A range in Excel refers to a group of contiguous cells selected together.
- ❖ It can represent rows, columns, or both, depending on the dataset layout.
- ❖ Example: The range A1:A10 represents cells from A1 to A10 in a single column.

❖ Naming a Range

- ❖ Select the range.
- ❖ Go to the Formulas tab and click Define Name. Enter a name (e.g., "ChargesData") and click OK.

❖ Using Named Ranges

- ❖ Named ranges can be used directly in Excel formulas.
- ❖ Example: =AVERAGE(ChargesData) computes the average of the named range ChargesData.

ABSOLUTE VS RELATIVE REFERENCES IN EXCEL

- ❖ A **relative reference** changes when a formula is copied to another cell. Excel adjusts the reference based on the relative position of the new cell.
- ❖ An **absolute reference** remains constant when a formula is copied, regardless of where it is pasted. Absolute references are indicated by a dollar sign (\$) before the column and/or row.
- ❖ Example:
 - ❖ If you have the formula `=A$1 + B1` in cell C1 and copy it to cell C2, it will remain `=A$1 + B2`.
- ❖ Syntax:
 - ❖ `$A$1`: Fixes both the column and row.
 - ❖ `$A1`: Fixes only the column.
 - ❖ `A$1`: Fixes only the row.

MEAN (AVERAGE)

- ❖ Definition: The sum of all values divided by the total number of values.
- ❖ Example: For 5, 8, 10, 12
- ❖ $\text{Mean} = \frac{5+8+10+12}{4} = 8.75$
- ❖ Limitation:
 - ❖ Sensitive to extreme values (outliers), which can skew the result significantly.
- ❖ Example: For 5, 8, 10, 100
- ❖ $\text{Mean} = \frac{5+8+10+100}{4} = 30.75$
 - ❖ The mean does not represent the center well due to the outlier (100).
- ❖ Excel Formula: =AVERAGE(range)

MEDIAN (MIDDLE VALUE)

- ❖ Definition: The middle value when the data is sorted in ascending order.
- ❖ For Odd Number of Values: The median is the exact middle value.
 - ❖ Example: For 5, 8, 10 the median is 8.
- ❖ For Even Number of Values: The median is the average of the two middle values.
 - ❖ Example: For 5, 8, 10, 12,
 - ❖ Median = $\frac{8+10}{2} = 9$
- ❖ Less sensitive to extreme values, but doesn't always provide detailed information about data distribution.
- ❖ Excel Formula: = MEDIAN(range)

MODE (MOST FREQUENT VALUE)

- ❖ Definition: The value that appears most frequently in the dataset.
- ❖ Example: For 5, 8, 8, 10, 12, the mode is 8 because it appears twice.
- ❖ Special Cases:
 - ❖ Bimodal: A dataset with two modes. Example: 5, 8, 8, 10, 10, 12 → Modes are 8 and 10.
 - ❖ Multimodal: A dataset with more than two modes.
 - ❖ No Mode: If no value repeats, the dataset has no mode.
- ❖ Limitation: Mode may not exist in some datasets or may not be unique, making it less useful for numerical data analysis.
- ❖ Excel Formula:
 - ❖ For a single mode: =MODE.SNGL(range)
 - ❖ For multiple modes: Use =MODE.MULT(range) (entered as an **array formula**).

WHAT IS AN ARRAY FORMULA?

- ❖ An array formula in Excel is a formula that can perform multiple calculations on one or more items in a range.
- ❖ Unlike regular formulas, array formulas can return either a single result or multiple results.
- ❖ They are often used when a standard Excel function cannot achieve the desired outcome in a single step.

EXCEL 2019 OR EARLIER

- ❖ Array Formulas in Older Versions of Excel
- ❖ In older versions (Excel 2019 and earlier), users needed to press Ctrl + Shift + Enter to enter an array formula.
- ❖ Excel would then surround the formula with curly braces {}, indicating that it was processed as an array formula.
- ❖ Editing or modifying array formulas required re-entering them with Ctrl + Shift + Enter.

EXCEL 2021 OR EXCEL 365

- ❖ Modern versions of Excel (Excel 365 and Excel 2021) introduced dynamic arrays, eliminating the need for pressing Ctrl + Shift + Enter.
- ❖ Now, simply pressing Enter works for array formulas, and Excel automatically expands the results into adjacent cells as needed.
- ❖ This change makes it easier and more intuitive to work with array formulas.
- ❖ **Spill Feature**
 - ❖ The dynamic array engine introduces the spill feature, where results automatically populate into adjacent cells when an array formula returns multiple values.
 - ❖ If there is already data in the adjacent cells that prevents the array from expanding, Excel displays a #SPILL! error.

EXCEL 2021 OR EXCEL 365

- ❖ Other Functions That Support Dynamic Arrays
- ❖ SORT(range) – Automatically sorts a range and spills the result.
- ❖ UNIQUE(range) – Returns unique values from a range.
- ❖ FILTER(range, condition) – Filters a range based on a specified condition.

RANGE, MINIMUM, AND MAXIMUM

- ❖ Range: The difference between the maximum and minimum values in the dataset.
- ❖ Formula: $\text{Range} = \text{Max}(\text{range}) - \text{Min}(\text{range})$

STANDARD DEVIATION AND SAMPLE VARIANCE

❖ Standard Deviation

- ❖ A measure of how spread out the data is around the mean.
- ❖ The higher the standard deviation, the more dispersed the data is.

❖ Formula: *Sample Standard Deviation* $= \sqrt{\frac{\sum(x_i - \bar{x})^2}{(n-1)}}$

❖ Excel Formula: =STDEV.S(range)

❖ Sample Variance: It is the square of the standard deviation.

❖ Formula: *Variance* $= \frac{\sum(x_i - \bar{x})^2}{(n-1)}$

❖ Excel Formula: =VAR.S(range)

STANDARD DEVIATION AND POPULATION VARIANCE

❖ Standard Deviation

- ❖ A measure of how spread out the data is around the mean.
- ❖ The higher the standard deviation, the more dispersed the data is.

❖ Formula: *Population Standard Deviation* $= \sqrt{\frac{\sum(x_i - \bar{x})^2}{(n)}}$

❖ Excel Formula: =STDEV.P(range)

❖ Population Variance: It is the square of the standard deviation.

❖ Formula: *Variance* $= \frac{\sum(x_i - \bar{x})^2}{(n)}$

❖ Excel Formula: =VAR.P(range)

COUNT

Values
25
50
Text1
75
100
150
Text2
200

Function	Formula	Description	Example (Based on the Dataset)	Output
COUNT	=COUNT(A1:A10)	Counts the number of numeric values in a specified range.	=COUNT(A1:A10)	6
COUNTA	=COUNTA(A1:A10)	Counts the number of non-empty cells in a specified range, including numbers, text, and errors.	=COUNTA(A1:A10)	8
COUNTBLANK	=COUNTBLANK(A1:A10)	Counts the number of empty cells in a specified range.	=COUNTBLANK(A1:A10)	2
COUNTIF	=COUNTIF(A1:A10, ">50")	Counts the number of cells that meet a specific condition.	=COUNTIF(A1:A10, ">50")	4
COUNTIFS	=COUNTIFS(A1:A10, ">50")	Counts the number of cells that meet multiple criteria across different ranges.	=COUNTIFS(A1:A10, ">50", A1:A10, "<200")	3

INTRODUCTION TO D-FUNCTIONS IN EXCEL

- ❖ D Functions in Excel are specialized functions used to perform calculations on data stored in a structured table format, often referred to as a database in Excel.
- ❖ These functions allow you to apply multiple criteria and return results from specific columns.
- ❖ They are powerful tools for analyzing subsets of data without needing complex formulas or manual filtering.
- ❖ Efficient for performing operations like counting, summing, averaging, finding maximum and minimum values, and calculating variance and standard deviation on large datasets with multiple conditions.

INTRODUCTION TO D-FUNCTIONS IN EXCEL

- ❖ All D Functions require three inputs:
 - ❖ Database: The entire table, including headers.
 - ❖ Field: The specific column to perform the calculation on.
 - ❖ Criteria: A range defining the conditions for selecting rows.
- ❖ Unlike basic Excel functions, D Functions can handle multiple conditions across different columns seamlessly.

INTRODUCTION TO D-FUNCTIONS IN EXCEL

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D FUNCTIONS

Function	Formula	Description
DCOUNT	=DCOUNT(A1:D11, "Age", F1:F2)	Counts the numeric values in the Age column for "IT" only.
DCOUNTA	=DCOUNTA(A1:D11, "Age", F1:F2)	Counts the non-empty cells (including blanks) in the Age column for "IT".
DSUM	=DSUM(A1:D11, "Score", F1:F2)	Sums the Score values for "IT".
DAVERAGE	=DAVERAGE(A1:D11, "Score", F1:F2)	Returns the average of Score for "IT".
DMAX	=DMAX(A1:D11, "Age", F1:F2)	Returns the maximum Age for "IT".
DMIN	=DMIN(A1:D11, "Age", F1:F2)	Returns the minimum Age for "IT".
DGET	=DGET(A1:D11, "Score", F1:F2)	Returns a single Score value for "IT" (returns an error if multiple values match).
DSTDEV	=DSTDEV(A1:D11, "Score", F1:F2)	Estimates the standard deviation of the sample Score for "IT".
DSTDEVP	=DSTDEVP(A1:D11, "Score", F1:F2)	Calculates the population standard deviation of Score for "IT".
DVAR	=DVAR(A1:D11, "Score", F1:F2)	Estimates the variance of the sample Score for "IT".
DVARP	=DVARP(A1:D11, "Score", F1:F2)	Calculates the population variance of Score for "IT".

INTRODUCTION TO QUARTILES AND IQR

- ❖ Quartiles divide a dataset into four equal parts, each containing 25% of the data.
 - ❖ Q1 (First Quartile): The 25th percentile – the value below which 25% of the data lies.
 - ❖ Q3 (Third Quartile): The 75th percentile – the value below which 75% of the data lies.
- ❖ Interquartile Range (IQR) is the range between the first and third quartiles. It measures the spread of the middle 50% of the data.
- ❖ Formula: $IQR = Q3 - Q1$

STEPS TO CALCULATE Q1, Q3, AND IQR

- ❖ 1. Sort the Data: Arrange the data in ascending order.
 - ❖ 2. Find Q1: Q1 is the median of the lower half of the dataset (excluding the overall median if the number of data points is odd).
 - ❖ 3. Find Q3: Q3 is the median of the upper half of the dataset.
 - ❖ 4. Calculate IQR: Subtract Q1 from Q3:
-
- ❖ $IQR = Q3 - Q1$

STEPS TO CALCULATE Q1, Q3, AND IQR

- ❖ Sorted Dataset: {5, 8, 12, 14, 18, 21, 24, 30, 35}
- ❖ Median (Overall): 18 (middle value)
- ❖ Q1: Median of the lower half {5, 8, 12, 14}: (Exclusive approach since 18 is excluded)
- ❖ $Q1 = (8 + 12) / 2 = 10$
- ❖ Q3: Median of the upper half {21, 24, 30, 35}: (Exclusive approach since 18 is excluded)
- ❖ $Q3 = (24 + 30) / 2 = 27$
- ❖ $IQR = Q3 - Q1 = 27 - 10 = 17$

STEPS TO CALCULATE Q1, Q3, AND IQR

- ❖ Sorted Dataset: {5, 8, 12, 14, 18, 21, 24, 30, 35}
- ❖ Median (Overall): 18 (middle value)
- ❖ Q1: Median of the lower half {5, 8, 12, 14, 18}: (Inclusive approach since 18 is excluded)
- ❖ $Q1 = 12$
- ❖ Q3: Median of the upper half {18, 21, 24, 30, 35}: (Exclusive approach since 18 is excluded)
- ❖ $Q3 = 24$
- ❖ $IQR = Q3 - Q1 = 24 - 12 = 12$

SKEWNESS

- ❖ Skewness describes the asymmetry in the data distribution.
- ❖ Types of Skewness:
- ❖ Symmetrical Distribution: $\text{Skewness} = 0$
- ❖ Positive Skew (Right-Skewed):
 - ❖ Longer tail on the right side, with most values concentrated on the left.
 - ❖ Example: Income distributions.
- ❖ Negative Skew (Left-Skewed):
 - ❖ Longer tail on the left side, with most values concentrated on the right.
 - ❖ Example: Age at retirement.

SKEWNESS

- Skewness Formula:

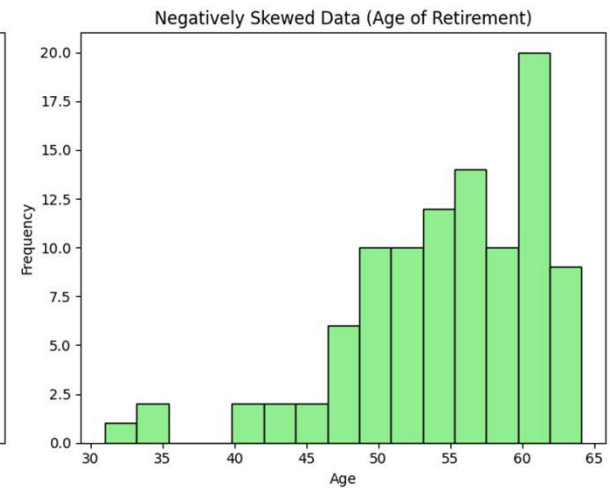
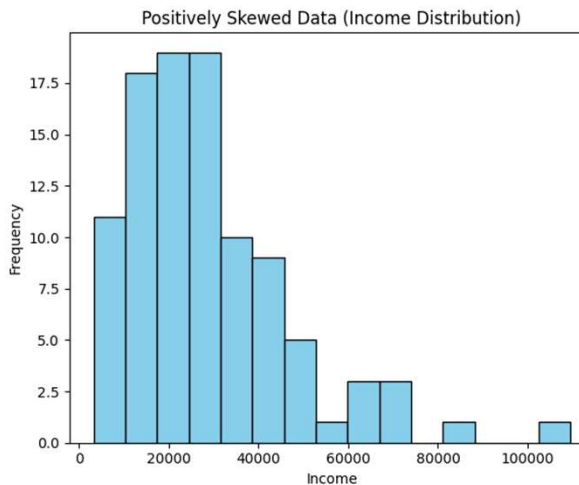
$$\text{Skewness} = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^3$$

- Where:

- x_i = data points
- $\bar{x}$ = mean
- s = standard deviation
- n = number of data points

- Interpretation:

- Positive Skew: Right-tailed.
- Negative Skew: Left-tailed.



SKEWNESS

- ❖ Skewness describes the asymmetry in the data distribution.
- ❖ Types of Skewness:
- ❖ Symmetrical Distribution: $\text{Skewness} = 0$
- ❖ Positive Skew (Right-Skewed):
 - ❖ Longer tail on the right side, with most values concentrated on the left.
 - ❖ Example: Income distributions.
- ❖ Negative Skew (Left-Skewed):
 - ❖ Longer tail on the left side, with most values concentrated on the right.
 - ❖ Example: Age at retirement.

SKEWNESS

❖ Practical Range for Normality:

- ❖ A skewness value between -0.5 and +0.5 is often considered approximately normal or symmetric in practice.
- ❖ Bell-shaped curve where $\text{mean} \approx \text{median} \approx \text{mode}$.

Skewness Value	Shape of Distribution	Interpretation
-0.5 to +0.5	Symmetrical	Approximate normal distribution
-1 to -0.5	Slightly Left-Skewed	Tail on the left
+0.5 to +1	Slightly Right-Skewed	Tail on the right
-1 to -2	Moderately Left-Skewed	More pronounced left tail
+1 to +2	Moderately Right-Skewed	More pronounced right tail
< -2 or > +2	Severely Skewed	Highly asymmetric, non-normal

KURTOSIS

- ❖ Kurtosis measures the tailedness of the data distribution.
- ❖ It tells us whether data have heavy tails (outliers) or light tails compared to a normal distribution.
- ❖ Types of Kurtosis:
 - ❖ Mesokurtic: Normal bell curve
 - ❖ Normal distribution kurtosis = 3.
 - ❖ Leptokurtic (Heavy tails): Tall, narrow peak with fat tails.
 - ❖ Kurtosis > 3 , more outliers.
 - ❖ Platykurtic (Light tails): Flatter peak with thin tails.
 - ❖ Kurtosis < 3 , fewer outliers.

KURTOSIS

- Kurtosis Formula:

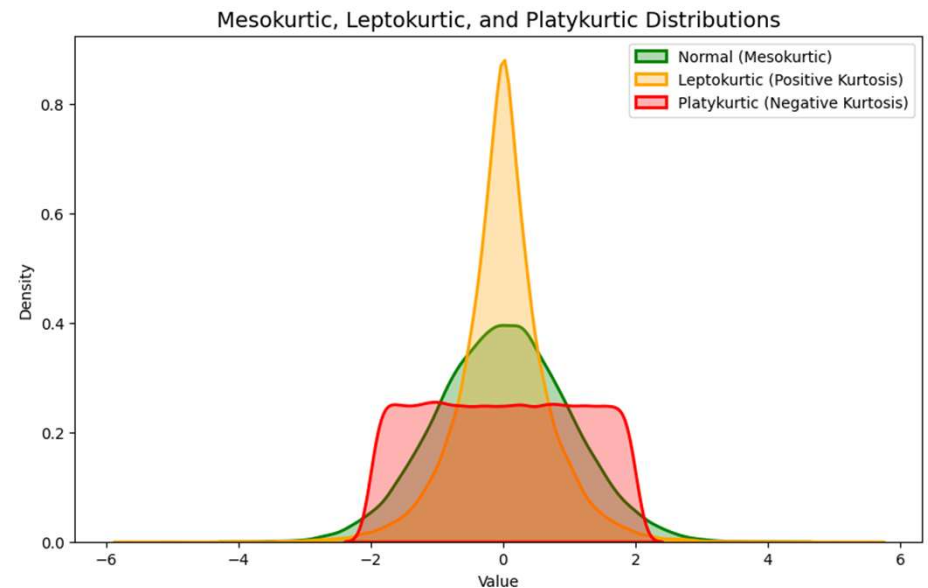
$$\text{Kurtosis} = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$$

- Interpretation:
 - Kurtosis > 3: **Leptokurtic** (Peaked distribution, heavy tails).
 - Kurtosis < 3: **Platykurtic** (Flatter distribution, light tails).

KURTOSIS

$$* \text{Excess Kurtosis} = \text{Kurtosis} - 3$$

- ❖ Kurtosis measures the tailedness of the data distribution.
- ❖ It tells us whether data have heavy tails (outliers) or light tails compared to a normal distribution.
- ❖ Types of Kurtosis:
 - ❖ Mesokurtic:
 - ❖ Normal distribution kurtosis = 3.
 - ❖ Leptokurtic (Tall Peak, Heavy tails):
 - ❖ Kurtosis > 3, more outliers.
 - ❖ Platykurtic (Flatter Peak, Light tails):
 - ❖ Kurtosis < 3, fewer outliers.



SKEWNESS AND KURTOSIS (EXCEL FUNCTIONS)

Function	Query	Example Formula	Interpretation
SKEW()	What is the skewness of the sales across all products and regions?	=SKEW(C2:C11)	A skewness value between - 0.5 and +0.5 is considered approximately symmetric. Positive values indicate right skew, and negative values indicate left skew.
KURT()	What is the kurtosis of the sales across all products and regions?	=KURT(C2:C11)	Kurtosis = 3 indicates a normal (Mesokurtic) distribution. Kurtosis > 3 (Leptokurtic) implies heavy tails, and Kurtosis < 3 (Platykurtic) implies light tails.